



ESP32-S3 Microphone & RGB LED Ring PCB

Six-Microphone PDM Array and RGB Status Ring

System Architecture & Preliminary Design

Technical Engineering Report

Project:	Microphone / RGB LED Ring PCB
Platform:	ESP32-S3 PoE Speaker Main Board
Interface:	J4 – 3.3 V, PDM, I2C and spare GPIO
Configuration:	6 digital PDM microphones + 6 RGB status LEDs
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Field:	Electronics / PCB Design / Embedded Hardware
Document:	System Architecture & Preliminary Design
Date:	September 2026
Revision:	1.1

Engineering design document for technical review. The principal active components and system architecture have been selected. Detailed electrical values, PCB layout implementation, mechanical integration and manufacturing details will be finalized during schematic capture and PCB design.

I. Project Description and Requirements

The objective of this project is to design a dedicated microphone and RGB status LED ring PCB for the existing ESP32-S3 PoE speaker main board.

The ring will be installed behind the speaker grille and will contain six digital PDM MEMS microphones and six RGB LEDs distributed around its circumference. The first mechanical implementation is intended for a speaker with an outside diameter of approximately 9 inches.

The ring will be connected to the existing main board through a single 14-pin connector, J4.

The existing interface provides a 3.3 V supply, ground connections, one PDM clock, four PDM data lines, I2C SDA/SCL and one spare GPIO.

The design objective is to maintain a reusable electrical architecture. Future speaker versions may change the ring diameter, component positions and mounting arrangement while preserving the same electrical interface and basic circuit architecture.

Table 1: Main ring PCB requirements

Parameter	Requirement
Main platform	Existing ESP32-S3 PoE speaker PCB
Microphones	6 digital PDM MEMS microphones
Indicators	6 low-brightness RGB status LEDs
Main connection	Single 14-pin J4 connector
Available supply	3.3 V from the existing main board
Digital interfaces	PDM, I2C and spare GPIO
Mechanical concept	Continuous ring PCB behind speaker grille
Initial speaker size	Approximately 9-inch outside diameter
Design objective	Reusable electrical architecture

II. Existing Main-Board Interface

The existing ESP32-S3 main-board schematic and PCB were reviewed to verify the electrical interface available for the new ring.

Connector J4 provides the power and digital signals required for the microphone and RGB ring.

Table 2: J4 interface and ESP32-S3 GPIO mapping

Pin	Signal	ESP32-S3 GPIO
1	3V3	—
2	GND	—
3	PDM_CLK	GPIO4
4	GND	—
5	PDM_DATA0	GPIO5
6	GND	—
7	PDM_DATA1	GPIO6
8	GND	—
9	PDM_DATA2	GPIO7
10	GND	—
11	PDM_DATA3	GPIO8
12	I2C_SDA	GPIO18
13	I2C_SCL	GPIO21
14	RING_GPIO_SPARE	GPIO1

The existing speaker audio interface uses separate standard-I2S signals: GPIO9 for BCLK, GPIO10 for LRCLK and GPIO11 for audio DATA.

For the ESP32-S3, hardware PDM-to-PCM receive conversion is supported on I2S0. Therefore, firmware shall reserve I2S0 for PDM microphone reception when hardware PDM-to-PCM conversion is used.

The existing standard-I2S audio output can be assigned to I2S1 when simultaneous microphone capture and audio playback are required.

The exact peripheral assignment remains a firmware configuration and shall be confirmed during firmware integration.

The 3.3 V supply available on J4 is generated on the main board by the TPS62132RGTR synchronous buck converter from the existing 5 V rail.

Before powering the new ring directly from J4, the available 3.3 V current margin must therefore be verified.

III. PDM Microphone Architecture

The six digital PDM microphones are organized as three shared-data pairs.

All six microphones receive the same PDM clock generated by the ESP32-S3:

$$\text{GPIO4} \rightarrow \text{PDM_CLK} \rightarrow 6 \text{ microphones}$$

The data organization is:

PDM_DATA0 : MIC1(L) + MIC2(R)

PDM_DATA1 : MIC3(L) + MIC4(R)

PDM_DATA2 : MIC5(L) + MIC6(R)

PDM_DATA3 : Reserved

This architecture uses three of the four PDM data lines available through J4. The fourth data line remains reserved for future expansion or design flexibility.

The selected microphone is the Infineon IM69D128S.

The device operates from approximately 1.62 V to 3.6 V and is therefore compatible with the 3.3 V rail provided by J4.

In Normal Mode, the supported PDM clock-frequency range is approximately 1.2 MHz to 3.3 MHz. The firmware PDM clock shall therefore be configured inside this operating range.

At a 3.072 MHz PDM clock, the microphone current consumption is approximately 520 μ A typical per microphone.

For six microphones:

$$I_{\text{MIC,total}} = 6 \times 0.520 = 3.12 \text{ mA}$$

The IM69D128S supports left/right channel selection, allowing two microphones to share one PDM data line while using the corresponding left/right timing relationship on the shared PDM stream.

Local supply decoupling shall be placed close to the VDD pad of each IM69D128S microphone. The Infineon documentation specifies local bypass decoupling for optimum microphone performance; the detailed implementation shall follow the final stereo-PDM application circuit and manufacturer layout guidance.

Where the selected datasheet/application configuration specifies a 100 nF local capacitor, this capacitor shall be placed directly at the microphone supply pins. Additional local bulk decoupling, including 1 μ F where appropriate, may be included according to the final power-distribution implementation.

This wording intentionally distinguishes the mandatory local high-frequency bypass function from any additional local bulk decoupling used as a PCB design choice.

The microphone acoustic-port orientation and footprint geometry shall be respected during PCB layout and aligned with the corresponding openings in the speaker grille.

IV. 3.3 V Power Budget

4.1 Purpose of the Calculation

The objective of this calculation is to determine whether the existing 3.3 V rail has sufficient current capability to supply the new microphone/RGB ring without requiring an additional local regulator.

The existing power path is:

$$5 \text{ V} \rightarrow \text{TPS62132RGTR} \rightarrow 3.3 \text{ V} \rightarrow \text{Main Board} + \text{J4 Ring}$$

The TPS62132RGTR provides a fixed 3.3 V output and has a nominal output-current capability of up to:

$$I_{\text{REG,rated}} = 3.0 \text{ A}$$

The output power corresponding to 3 A at 3.3 V is:

$$P_{\text{REG}} = V_{\text{OUT}} \times I_{\text{REG,rated}}$$

$$P_{\text{REG}} = 3.3 \times 3.0 = 9.9 \text{ W}$$

The 3 A device rating is not treated as guaranteed available current for the ring.

Actual system capability also depends on the upstream 5 V rail, inductor selection, PCB thermal performance, converter current limits and the existing main-board load.

4.2 Existing 3.3 V Loads

The principal active loads identified on the existing 3.3 V rail are:

- ESP32-S3-WROOM module;
- W5500 Ethernet controller;
- PCM5122 audio DAC;
- PCAL6416A I/O expander;
- pull-up and small bias networks.

Decoupling capacitors are not treated as continuous DC loads.

The 25 MHz crystal is part of the W5500 oscillator circuit and is not considered an independent 3.3 V load.

Protection devices are also not considered significant continuous loads under normal operating conditions.

Table 3: Existing 3.3 V current engineering budget

Load	Current	Basis
ESP32-S3-WROOM	355 mA	Wi-Fi transmit peak
W5500	132 mA	100-Mbit/s transmit operating value (typ.)
PCM5122	30 mA	Engineering allowance
PCAL6416A	1 mA	Engineering allowance
Pull-ups / small loads	10 mA	Conservative engineering allowance
Total	528 mA	Existing-board engineering budget

The values above represent the principal identified 3.3 V loads and are used as a conservative architecture-level budget.

Final production validation shall also confirm any additional minor loads present on the complete main-board schematic.

4.3 Main-Board Current Calculation

$$I_{\text{MAIN}} = I_{\text{ESP32}} + I_{\text{W5500}} + I_{\text{PCM5122}} + I_{\text{PCAL6416A}} + I_{\text{misc}}$$

$$I_{\text{MAIN}} = 355 + 132 + 30 + 1 + 10$$

$$I_{\text{MAIN}} = 528 \text{ mA}$$

The 528 mA value is an engineering budget rather than a measured worst-case current.

It combines a peak value, a typical operating value and conservative allowances for smaller loads.

4.4 Design Margin

A 25% engineering margin is applied:

$$I_{\text{MAIN,design}} = 1.25 \times I_{\text{MAIN}}$$

$$I_{\text{MAIN,design}} = 1.25 \times 528 = 660 \text{ mA}$$

The corresponding main-board power budget is:

$$P_{\text{MAIN,design}} = 3.3 \times 0.660 = 2.18 \text{ W}$$

The theoretical regulator current capability remaining before adding the ring is:

$$I_{\text{available}} = 3.00 - 0.660 = 2.34 \text{ A}$$

4.5 Ring Current Calculation

The selected Kingbright APHF1608LSEEQBDZGKC is intended for low-current RGB status indication.

For the preliminary power calculation, each of the 18 independently controlled color channels is budgeted at 2 mA.

The maximum simultaneous LED-channel budget used for this calculation is therefore:

$$I_{\text{LEDs}} = 6 \times 3 \times 2 \text{ mA}$$

$$I_{\text{LEDs}} = 36 \text{ mA}$$

The six IM69D128S microphones consume approximately:

$$I_{\text{MIC,total}} = 6 \times 0.520 = 3.12 \text{ mA}$$

The LP5018 internal consumption is small compared with the LED-channel load.

A conservative 1 mA architecture-level allowance is included for LED-driver internal consumption and associated small control-current overhead.

Therefore:

$$I_{\text{RING,estimated}} = I_{\text{LEDs}} + I_{\text{MIC,total}} + I_{\text{driver}}$$

$$I_{\text{RING,estimated}} = 36 + 3.12 + 1 = 40.12 \text{ mA}$$

A 25% design margin is then applied to the calculated ring load:

$$I_{\text{RING,design}} = 1.25 \times 40.12 = 50.15 \text{ mA}$$

For practical design budgeting:

$$I_{\text{RING,design}} \approx 51 \text{ mA}$$

The corresponding ring power budget is:

$$P_{\text{RING,design}} = 3.3 \times 0.051 \approx 0.168 \text{ W}$$

The previous 150 mA architecture allocation may still be regarded as a conservative upper design reserve, but it is not the expected ring operating current.

4.6 Total 3.3 V Budget

Using the component-level ring design budget:

$$I_{\text{TOTAL}} = I_{\text{MAIN,design}} + I_{\text{RING,design}}$$

$$I_{\text{TOTAL}} = 660 + 51 = 711 \text{ mA}$$

The corresponding total 3.3 V power is:

$$P_{\text{TOTAL}} = 3.3 \times 0.711 \approx 2.35 \text{ W}$$

The regulator utilization relative to the 3 A device rating is:

$$\text{Utilization} = \frac{0.711}{3.00} \times 100 \approx 23.7\%$$

The theoretical remaining current headroom is:

$$I_{\text{HEADROOM}} = 3.00 - 0.711 = 2.289 \text{ A}$$

4.7 Power-Budget Summary

Table 4: 3.3 V architecture-level power-budget summary

Parameter	Calculated Value
TPS62132 rated output current	3.00 A
Output power corresponding to 3 A at 3.3 V	9.90 W
Existing main-board engineering budget	528 mA
Existing main-board budget with 25% margin	660 mA
Calculated ring current before margin	40.12 mA
Ring design current with 25% margin	≈ 51 mA
Combined 3.3 V design budget	711 mA
Combined 3.3 V power	≈ 2.35 W
Regulator utilization relative to 3 A rating	$\approx 23.7\%$
Theoretical current headroom	≈ 2.29 A

4.8 Power-Budget Conclusion

The architecture-level calculation indicates that the existing TPS62132RGTR has substantial rated-current margin for the microphone and RGB LED ring.

The selected ring components result in an estimated current of approximately 40.12 mA when all 18 LED channels are simultaneously budgeted at 2 mA.

After applying a 25% design margin, approximately 51 mA is allocated to the ring.

Combined with the 660 mA main-board design budget, the total 3.3 V design budget is approximately 711 mA, corresponding to about 23.7% of the TPS62132 3 A device rating.

Therefore, an additional 3.3 V regulator is not justified by the architecture-level current requirement, and the ring can be supplied from the existing J4 3.3 V rail.

Final electrical sign-off shall nevertheless verify the complete main-board load, upstream 5 V capability, converter thermal conditions, inductor rating and PCB implementation.

V. System Architecture

The existing ESP32-S3 main board remains the system controller.

The ring PCB connects to the existing J4 interface and contains the six-microphone PDM array and RGB status-indication circuitry.

Figure 1 presents the functional architecture.

Detailed signal-level connections are listed separately in Table 5.

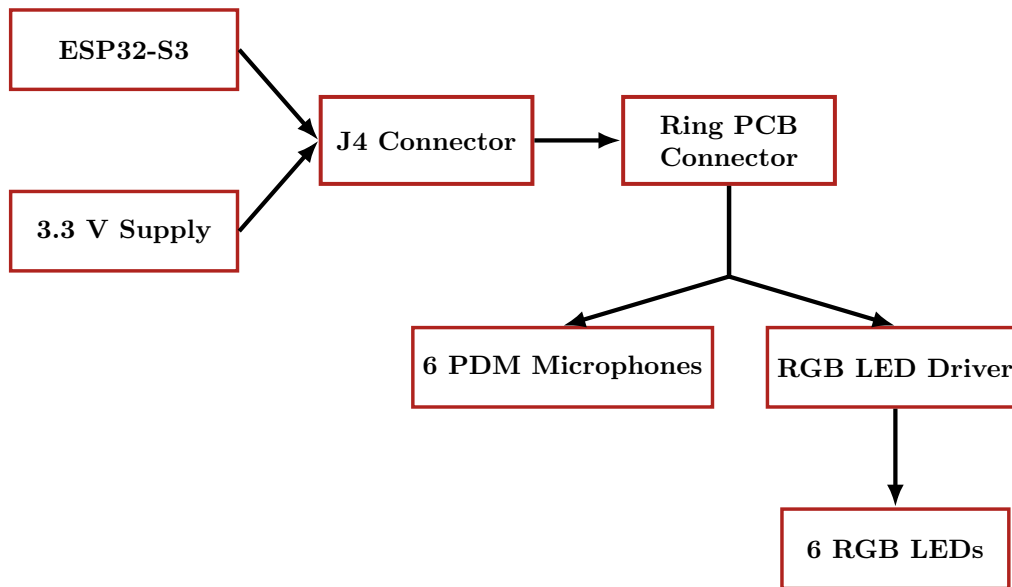


Figure 1: Simplified system architecture showing the interface between the existing ESP32-S3 main PCB and the microphone/RGB LED ring PCB.

5.1 Main-Board to Ring Connections

The detailed electrical interface between the existing main PCB and the new ring is summarized in Table 5.

Table 5: Electrical connections between the existing main PCB and the microphone/RGB LED ring

J4 Pin	Signal	ESP32-S3	Ring Connection / Function
1	3V3	–	3.3 V supply for microphone and LED-control circuitry
2	GND	–	Ground return
3	PDM_CLK	GPIO4	Common PDM clock distributed to all six microphones
4	GND	–	Ground return
5	PDM_DATA0	GPIO5	Shared PDM data from MIC1 and MIC2
6	GND	–	Ground return
7	PDM_DATA1	GPIO6	Shared PDM data from MIC3 and MIC4
8	GND	–	Ground return
9	PDM_DATA2	GPIO7	Shared PDM data from MIC5 and MIC6
10	GND	–	Ground return
11	PDM_DATA3	GPIO8	Reserved for future expansion; no functional load in the present design
12	I2C_SDA	GPIO18	I2C data connection to RGB LED driver
13	I2C_SCL	GPIO21	I2C clock connection to RGB LED driver
14	RING_GPIO_SPARE	GPIO1	Reserved control signal; candidate for LP5018 enable control

Ring PCB Component Selection

The principal active components selected for the ring PCB are summarized in Table 6.

Table 6: Main component selection for the microphone/RGB LED ring PCB

Reference	Function	Selected Component		Qty.	Design Role
MIC1– MIC6	Digital PDM microphone	Infineon IM69D128S		6	Digital MEMS microphones organized as three shared-data PDM pairs using a common PDM clock.
U1	RGB LED driver	Texas LP5018	Instruments	1	18-channel I2C constant-current sink driver providing independent control of the 18 LED color channels.
LED1– LED6	RGB status LED	Kingbright APHF1608LSEEQBDZGKC		6	Low-current RGB LEDs intended for status-indication operation at a preliminary 2 mA per color-channel design point.

5.2 PDM Microphone Block

The six Infineon IM69D128S microphones are arranged as three shared-data PDM pairs:

DATA0 : MIC1 + MIC2

DATA1 : MIC3 + MIC4

DATA2 : MIC5 + MIC6

All six microphones share the same PDM clock.

Each microphone pair uses the device left/right channel-selection function so that two microphones can share one PDM data line.

The PDM clock will be configured within the IM69D128S Normal Mode range of approximately 1.2 MHz to 3.3 MHz.

Each microphone will receive local high-frequency supply bypass decoupling placed directly adjacent to its VDD pad.

The exact capacitor implementation shall follow the final IM69D128S stereo-PDM application configuration and manufacturer layout guidance.

A 100 nF local bypass capacitor may be used where specified by the selected application circuit, while additional local bulk decoupling such as 1 μ F may be included where justified by the final power-distribution design.

PDM_DATA3 remains unused by the present six-microphone architecture.

It may be routed to an accessible test pad while remaining electrically unloaded during normal operation.

Microphone placement shall maintain equal angular distribution around the ring as far as permitted by the final mechanical design.

Acoustic ports shall be aligned with suitable openings through the speaker grille.

5.3 RGB LED Control Block

The ring contains six Kingbright APHF1608LSEEQBDZGKC low-current RGB status LEDs.

Each RGB LED contains three independently controlled color channels.

The total required number of LED channels is therefore:

$$N_{\text{LED channels}} = 6 \times 3 = 18$$

The Texas Instruments LP5018 provides 18 constant-current sink outputs and is controlled from the ESP32-S3 through the existing I2C interface.

The control path is:

$$\text{ESP32-S3} \rightarrow \text{I2C} \rightarrow \text{LP5018} \rightarrow 18 \text{ LED channels}$$

The selected RGB LED will be operated as a low-current status indicator, with a preliminary design point of approximately 2 mA per active color channel.

The LED polarity and exact pin mapping shall be verified directly against the selected Kingbright ordering-code datasheet before final schematic release.

Because the LP5018 is a current-sink LED driver, the final LED connection must be electrically compatible with sink-current drive.

The exact common-terminal configuration of the selected RGB LED shall therefore be confirmed during schematic capture rather than inferred from package naming alone.

At the preliminary 2 mA operating point, LED forward-voltage characteristics shall be verified against the manufacturer's electrical characteristics.

Particular attention shall be given to the blue and green channels, because their forward voltage can approach the 3.3 V supply.

The available LED-driver compliance margin is approximately:

$$V_{\text{HEADROOM}} = V_{\text{RING}} - V_F$$

For example, if a color channel reaches:

$$V_F = 3.1 \text{ V}$$

while the ring supply is:

$$V_{\text{RING}} = 3.3 \text{ V}$$

then the nominal remaining voltage is only:

$$V_{\text{HEADROOM}} = 3.3 - 3.1 = 0.2 \text{ V}$$

before accounting for supply tolerance and PCB/interconnect voltage drop.

The final schematic shall therefore verify the LP5018 output compliance requirement against worst-case LED forward voltage and minimum ring supply voltage.

The LP5018 uses integrated PWM brightness control.

Its switching activity shall therefore be treated as a potential noise-coupling source near the sensitive microphone circuitry.

LED-current return paths, driver decoupling and microphone supply/ground routing shall be arranged to minimize coupling between the LED and microphone sections.

5.4 I2C and Control Configuration

The LP5018 includes hardware address-selection capability.

Its final address configuration shall be selected after confirming the addresses of the other devices connected to the existing I2C bus.

The existing I2C pull-up network on the main board shall also be verified before additional pull-up resistors are placed on the ring.

Duplicate pull-up networks shall be avoided unless the resulting equivalent resistance is intentionally required by the final bus design.

The LP5018 EN input shall not be left floating.

RING_GPIO_SPARE (GPIO1) is available as a candidate for software control of the LP5018 enable input.

If GPIO1 is used for this purpose, a defined hardware default state shall be provided so that the LED driver remains in a known state during reset and boot.

A pull-down network may be used if the final GPIO1 boot-state analysis and LP5018 EN thresholds confirm that this is the desired implementation.

The exact resistor value shall be selected during schematic capture rather than assumed at architecture stage.

Alternatively, the EN pin may be assigned a fixed hardware state if software enable control is not required.

The final implementation will be selected during schematic capture after verifying GPIO1 boot behavior and the desired system startup sequence.

5.5 Reserved Interface

PDM_DATA3, connected to ESP32-S3 GPIO8, is not required for the present six-microphone architecture and remains reserved for future expansion.

An accessible test point may be provided on this signal without loading the line.

RING_GPIO_SPARE, connected to ESP32-S3 GPIO1, is reserved as a general ring-control signal and may be used for LP5018 enable control subject to the startup-state verification described above.

VI. PCB Implementation Considerations

The ring PCB combines sensitive digital MEMS microphones with PWM-controlled LED-current paths.

PCB implementation shall therefore maintain controlled power distribution and low-impedance return-current paths.

The following layout principles will be applied:

- Keep microphone local high-frequency bypass capacitors directly adjacent to the corresponding IM69D128S VDD pads.
- Provide additional local supply decoupling only where justified by the final microphone power-distribution implementation and manufacturer guidance.
- Maintain a continuous low-impedance ground reference around the ring.
- Keep LED-current return paths and LP5018 switching-current loops away from microphone supply and data routing where practical.
- Provide adequate local decoupling at the LP5018.
- Route the common PDM clock cleanly and avoid unnecessary stubs.
- Provide an optional series-damping footprint near the PDM clock source or ring entry if signal-integrity verification indicates that damping is required.
- Verify 3.3 V and ground voltage drop around the approximately 9-inch ring using the final copper thickness, trace/plane width and operating current.
- Use sufficiently wide power distribution rather than assuming that a specific PCB layer count is required solely from the ring diameter.
- Maintain microphone acoustic-port clearance according to the manufacturer footprint recommendations.
- Keep the LP5018 and LED-current routing physically separated from the most sensitive microphone supply and PDM routing where practical.
- Avoid routing fast digital or LED-current traces immediately adjacent to microphone acoustic-port regions.

Because the LP5018 uses PWM brightness control, coupling of LED switching activity into the microphone supply and ground network will be considered during layout.

Additional filtering, such as a ferrite-isolated LED-driver supply branch with local capacitance, may be evaluated if layout analysis or prototype measurements indicate that further isolation is necessary.

Such filtering is therefore considered a design option rather than a mandatory architecture requirement at this stage.

VII. Items to Verify Before Schematic and PCB Release

The principal architecture and active-component selection are defined.

The following implementation details remain to be confirmed during schematic capture and PCB development:

- Final ESP32-S3 PDM clock setting within the IM69D128S operating range;
- Exact IM69D128S package footprint, pad geometry and acoustic-port orientation;
- Final IM69D128S local bypass network based on the selected stereo-PDM application configuration and manufacturer guidance;
- Confirmation of simultaneous firmware use of I2S0 for hardware PDM reception and the required standard-I2S peripheral for speaker playback;
- LP5018 EN implementation and GPIO1 startup/boot behavior;
- Final hardware default state for LP5018 EN during ESP32-S3 reset and startup;
- LP5018 hardware I2C address configuration and compatibility with the existing bus devices;
- Existing I2C pull-up network and resulting total bus pull-up resistance;
- Exact Kingbright APHF1608LSEEQBDZGKC polarity and pin mapping;
- Electrical compatibility between the selected RGB LED topology and the LP5018 current-sink outputs;
- LP5018 output-voltage compliance/headroom with the selected RGB LED at worst-case forward voltage and minimum 3.3 V ring supply;
- Final programmed RGB LED channel current;
- Final component-level ring current after the LED-current setting is frozen;
- LP5018 local decoupling and optional supply-isolation requirement;
- Potential coupling of LP5018 PWM switching activity into the microphone supply, ground and digital signal network;
- Need for an optional ferrite-isolated LED-driver supply branch based on layout analysis and prototype measurements;
- PDM clock series-damping requirement based on routing geometry and signal-integrity behavior;
- 3.3 V and ground voltage drop around the final ring geometry;
- Complete main-board 3.3 V load verification, including minor loads not included in the principal-load budget;

- TPS62132 implementation margin including the upstream 5 V rail, inductor rating, thermal conditions and PCB layout;
- Ring outside diameter and inside diameter;
- Final mounting-hole positions and dimensions;
- Final component side and microphone acoustic-port orientation relative to the speaker grille;
- Exact mating connector and cable implementation for J4;
- Cable length between the main PCB and ring PCB, particularly for PDM clock signal-integrity verification;
- Microphone acoustic-port alignment with the final speaker grille;
- Final production footprints and manufacturer sourcing references.

VIII. Preliminary Design Conclusion

The preliminary architecture supports a six-microphone digital PDM array and six RGB status LEDs using the existing J4 interface of the ESP32-S3 speaker main board.

The six microphones are organized as three shared-data PDM pairs using PDM_DATA0, PDM_DATA1 and PDM_DATA2, while PDM_DATA3 remains available for future expansion or test access.

The selected Infineon IM69D128S microphone is electrically compatible with the available 3.3 V supply and supports the required shared-data PDM architecture.

The microphone local supply-decoupling network will be finalized from the selected manufacturer application configuration during schematic capture, with high-frequency bypass capacitors placed immediately adjacent to the microphone supply pins.

The Texas Instruments LP5018 provides the required 18 current-sink channels for independent control of the six RGB LEDs through the existing I2C interface.

Final schematic verification will confirm LED polarity, output-voltage headroom, I2C addressing, bus pull-ups and the LP5018 enable strategy.

The calculated ring current is approximately 40.12 mA before design margin and approximately 51 mA after applying a 25% engineering margin.

The combined architecture-level 3.3 V design budget is approximately 711 mA.

This indicates that an additional local 3.3 V regulator is not required based on current demand alone.

Final electrical sign-off remains dependent on confirmation of the complete main-board load, upstream 5 V capability, converter implementation, voltage drop, thermal conditions and final PCB layout.

The architecture is therefore considered suitable to proceed to detailed schematic capture once the remaining mechanical and interface details are confirmed.

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